Neural Plate: The Beginning of the Nervous System

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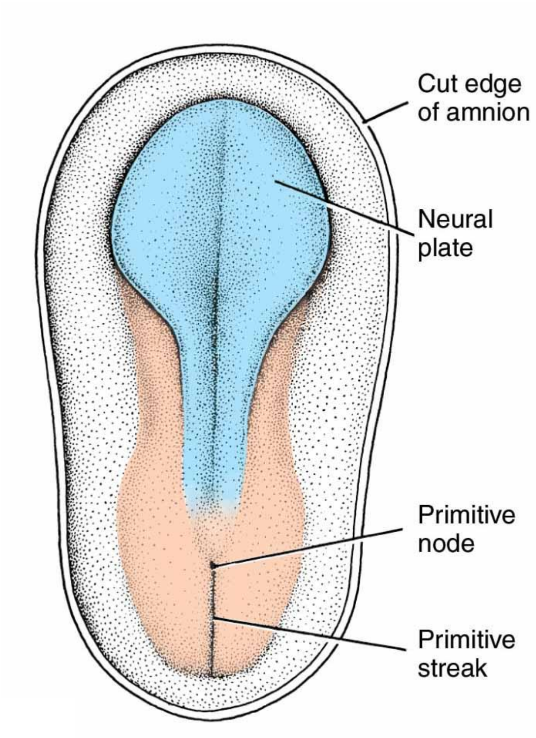

19 Days

The notochord and paraxial mesenchyme induces the overlying ectoderm to thicken into the neural plate.

This begins rostrally and progresses caudally.

Involves many signaling molecules, including:
- transforming growth factor-β
- Wnts
- sonic hedgehog (SHH)
- bone morphogenic proteins (BMPs)

NEURULATION

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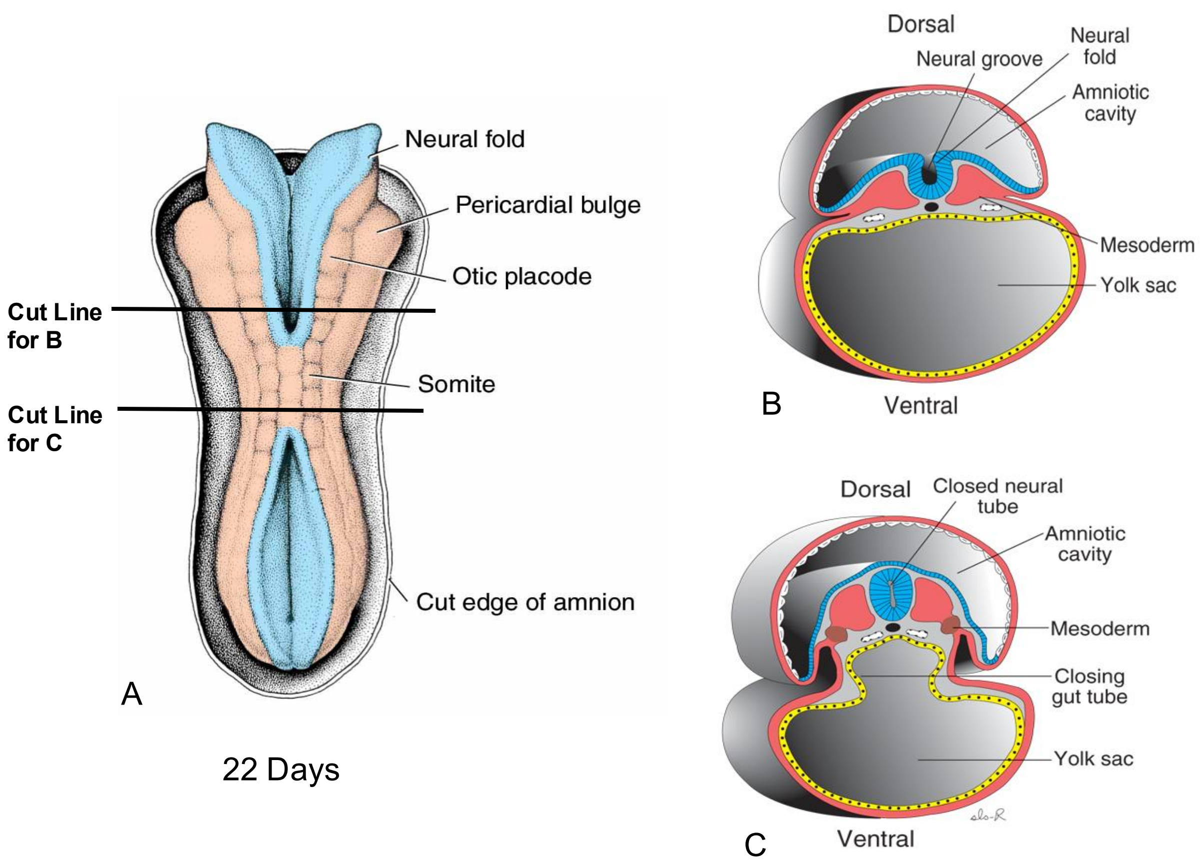

3-2C Neurulation: Neural tube closure

Neurulation (Cross-Sectional View)

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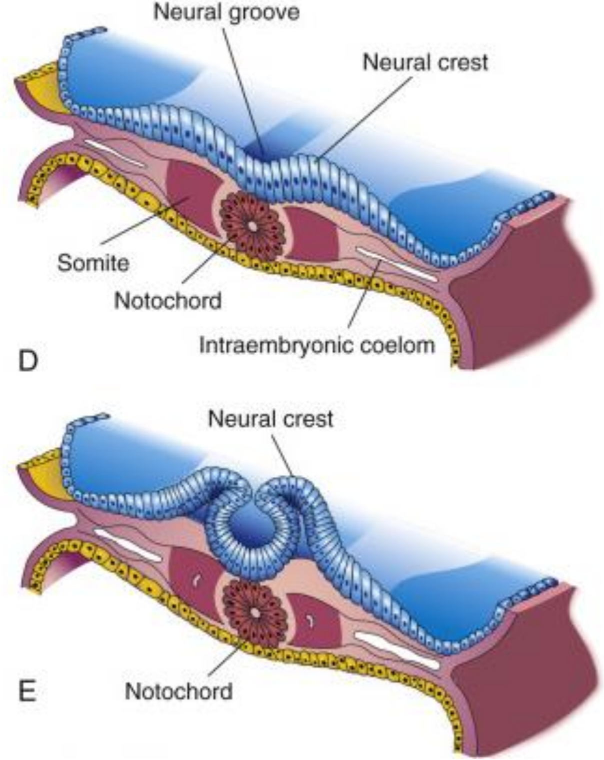
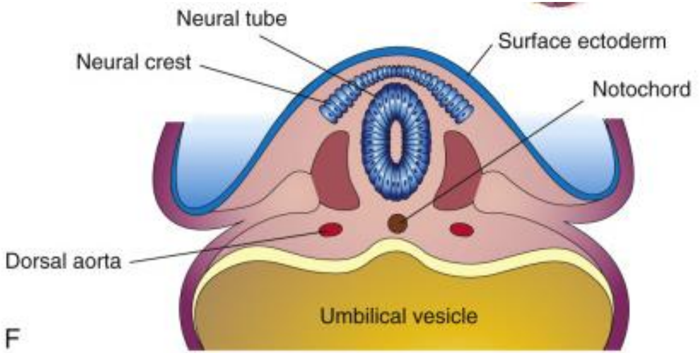
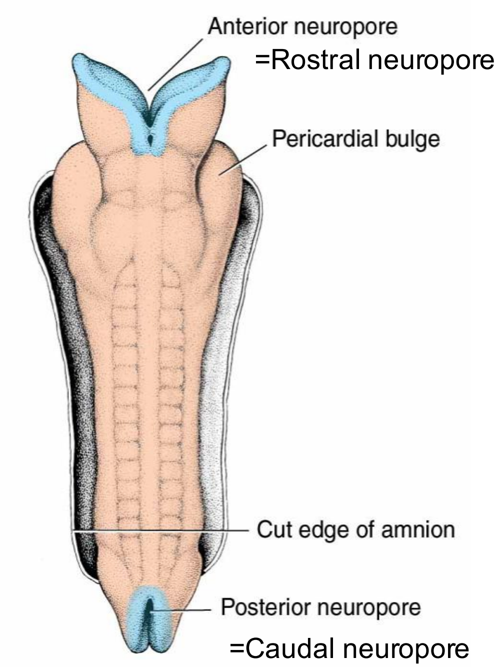




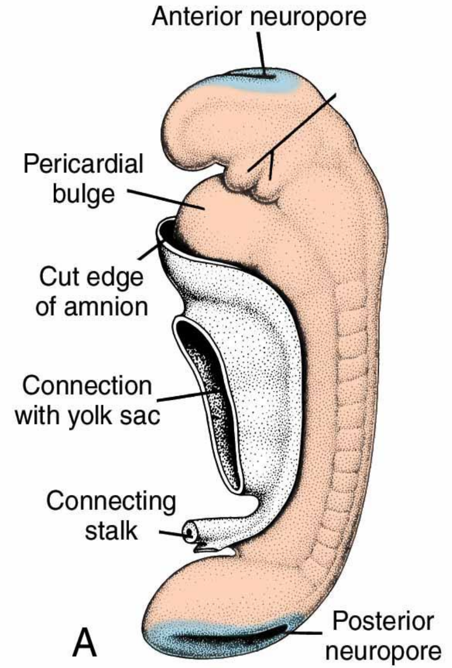
Sections from different levels at 22 days development.

Moore, The Developing Human, Fig. 17.1

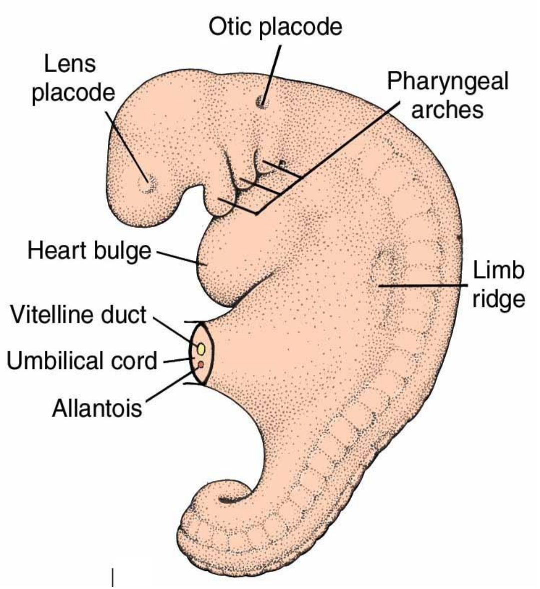
NEURULATION



23 Days



25 Days

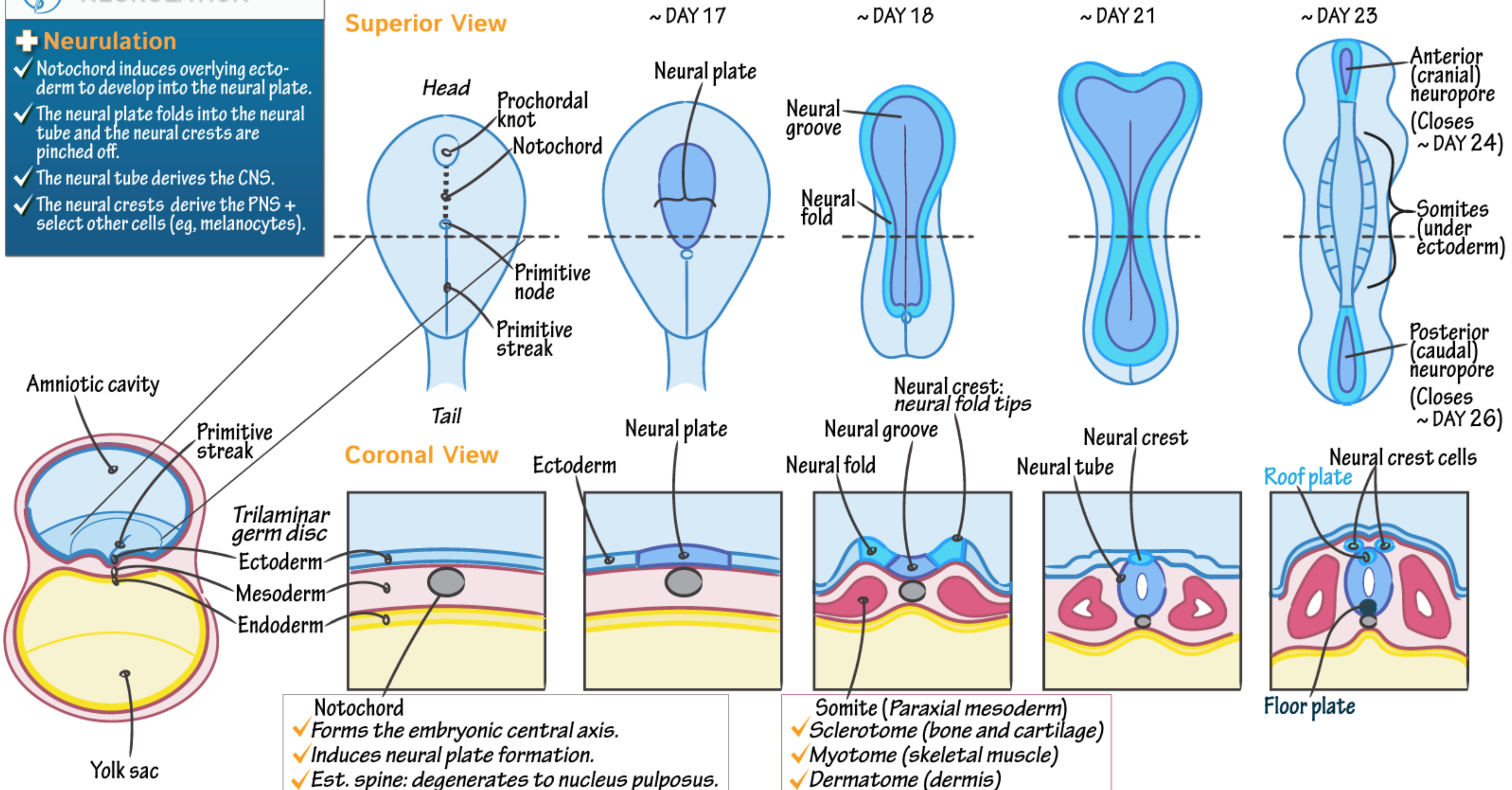


28 Days

NEURULATION

Neurulation

- ✓ Notochord induces overlying ectoderm to develop into the neural plate.
- ✓ The neural plate folds into the neural tube and the neural crests are pinched off.
- ✓ The neural tube derives the CNS.
- ✓ The neural crests derive the PNS + select other cells (eg, melanocytes).



Notochord
- ✓ Forms the embryonic central axis.
- ✓ Induces neural plate formation.
- ✓ Est. spine: degenerates to nucleus pulposus.

Somite (Paraxial mesoderm)
- ✓ Sclerotome (bone and cartilage)
- ✓ Myotome (skeletal muscle)
- ✓ Dermatome (dermis)

Failure of Caudal Neuropore Closure: Spina Bifida

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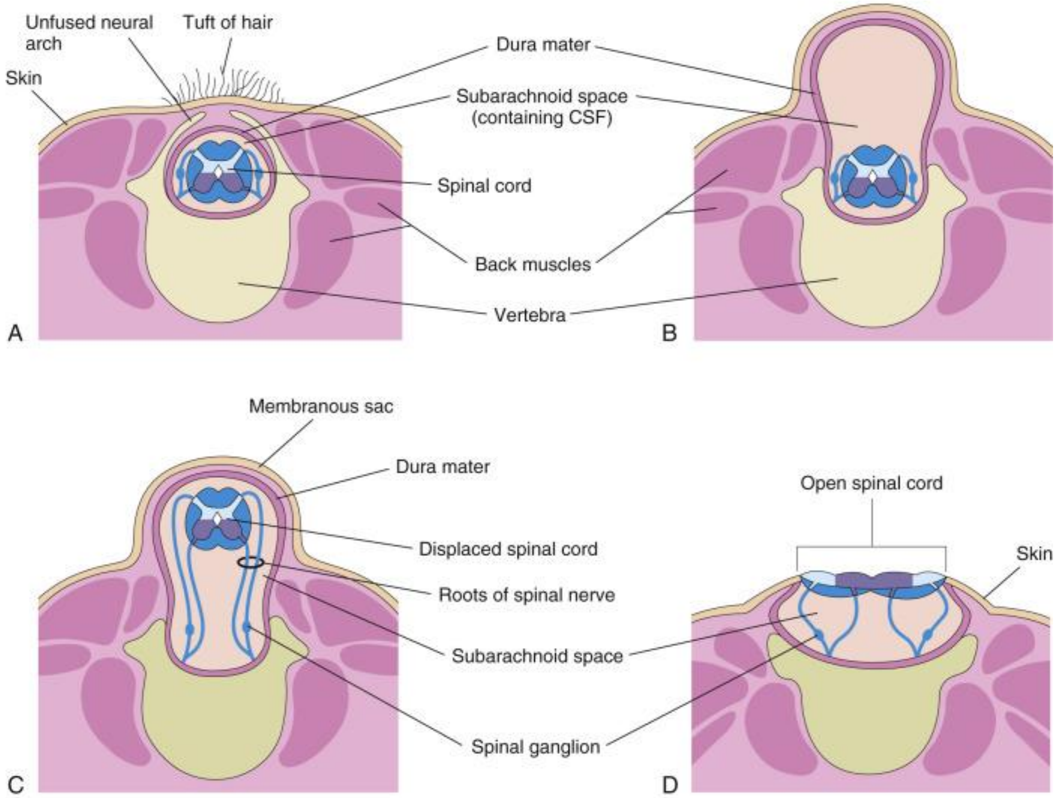

Moore, The Developing Human, Fig. 17.12

A. Spina bifida occulta—often found incidentally with no symptoms
B. Meningocele
C. Meningomyelocele (myelomeningocele)
D. Myeloschisis

Failure of Rostral Neuropore Closure

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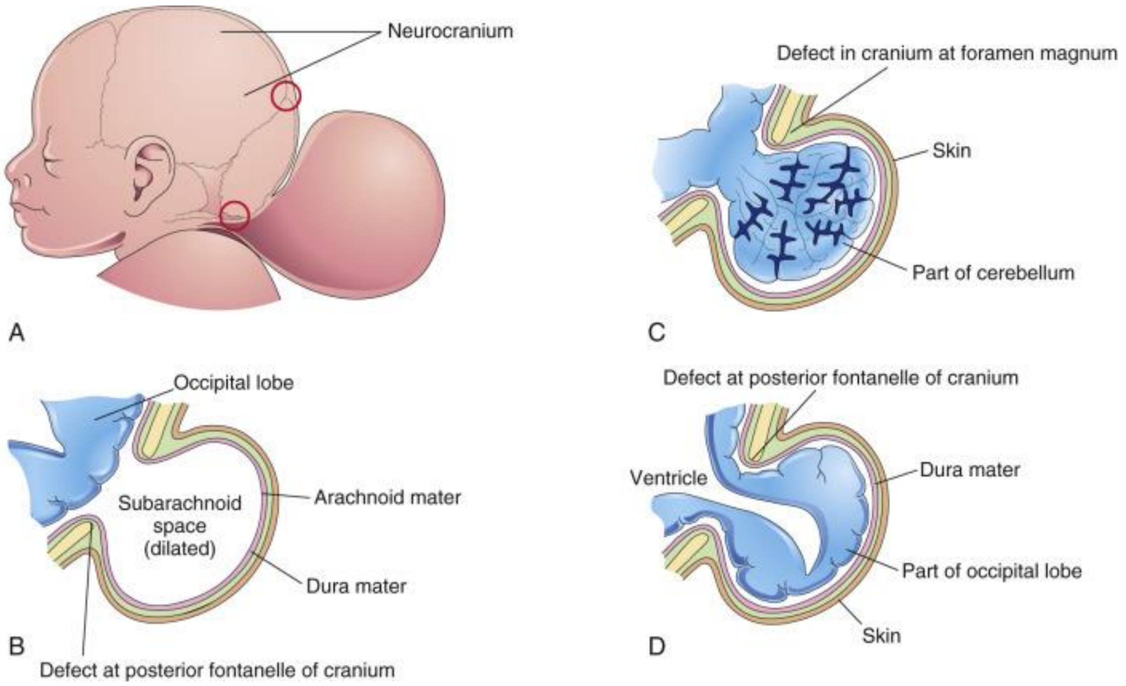


Encephalocele (cranium bifidum)—
a herniation of cranial contents (most common in occipital region)

Different degrees depending on herniation of neural tissue:

B: Meningocele

C: Meningoencephalocele

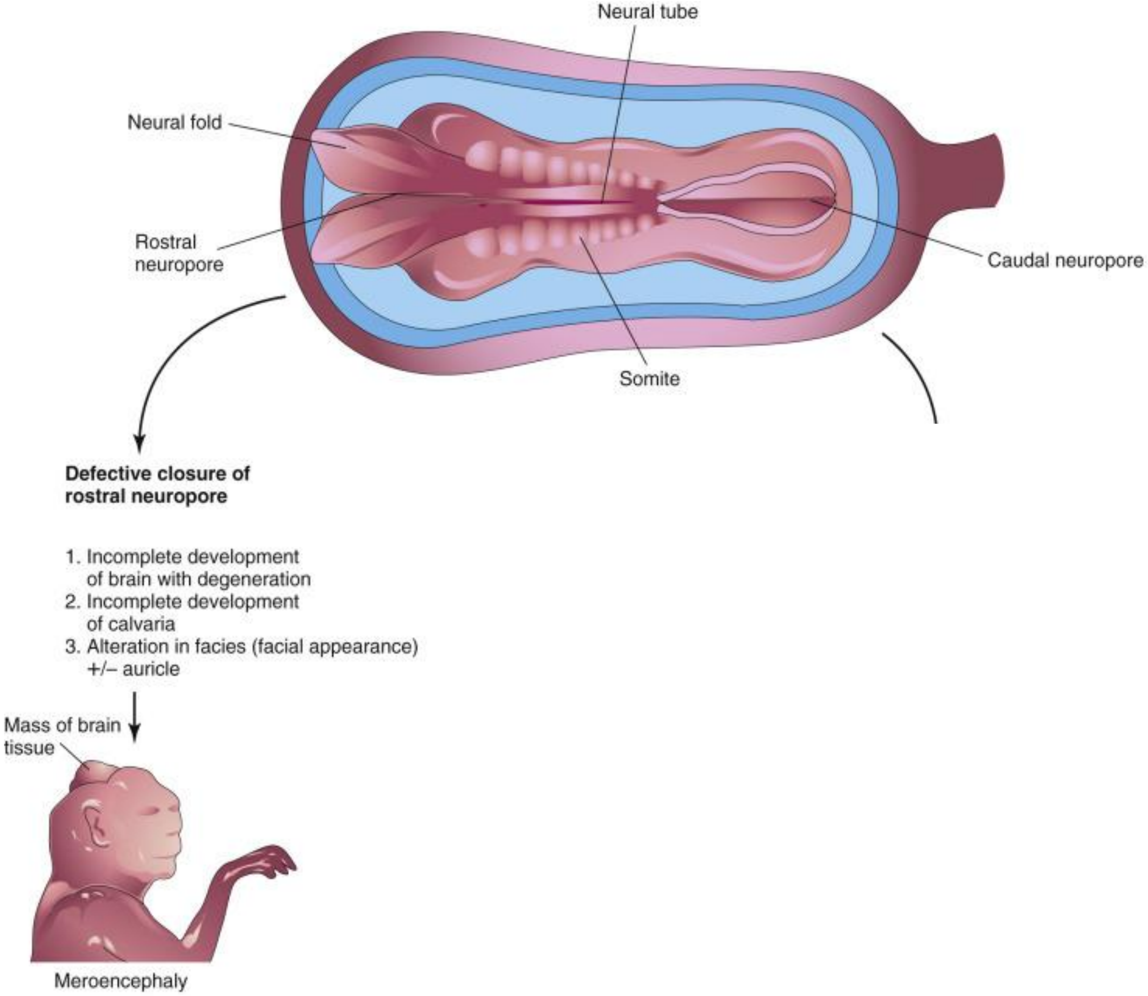
D: Meningohydroencephalocele

Moore, The Developing Human, Fig. 17.31

FAILURE OF ROSTRAL NEUROPORE CLOSURE

The most severe cases result in anencephaly (also called meroencephaly). From this point, the forebrain and overlying skull fail to develop.

This is the equivalent of myeloschisis for spina bifida, with exposed neural tissue.

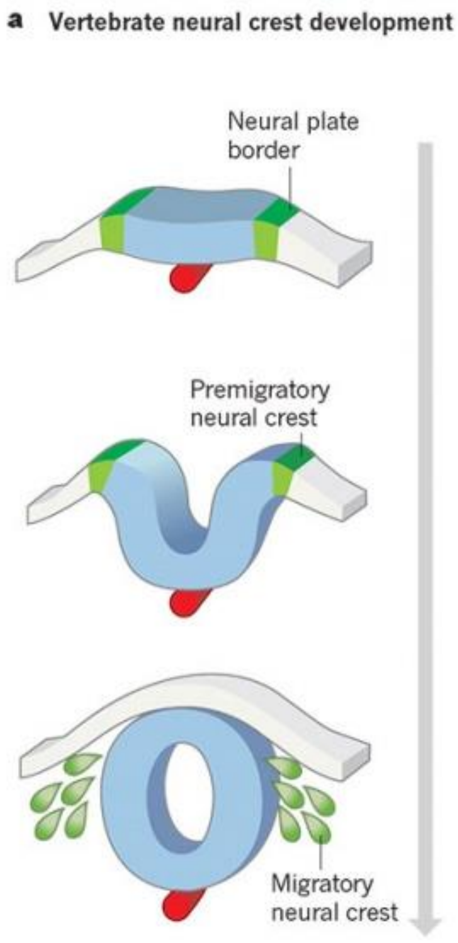


Moore, The Developing Human, Fig. 17.17

NEURAL CREST CELLS

Originate as ectoderm along the edge of the neural groove (that is—the neural crest)

As the neural tube separates from the external portion of ectoderm, neural crest cells separate from the ectoderm and migrate to various locations…



Green SA, Simoes-Costa M, Bronner ME. Nature. 2015 Apr 23;520(7548):474-82.

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NEURAL CREST CELLS

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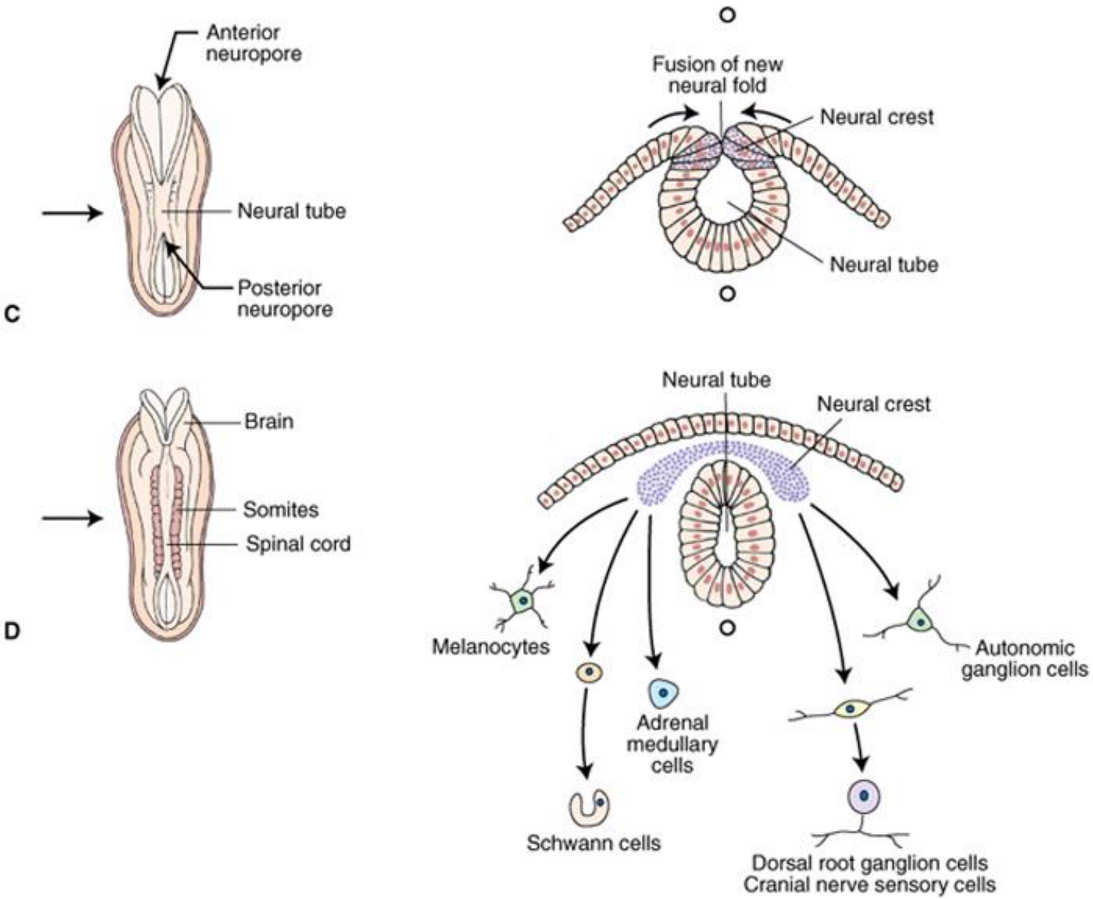


Siegel and Sapru, Essential Neuroscience, Fig. 2.1

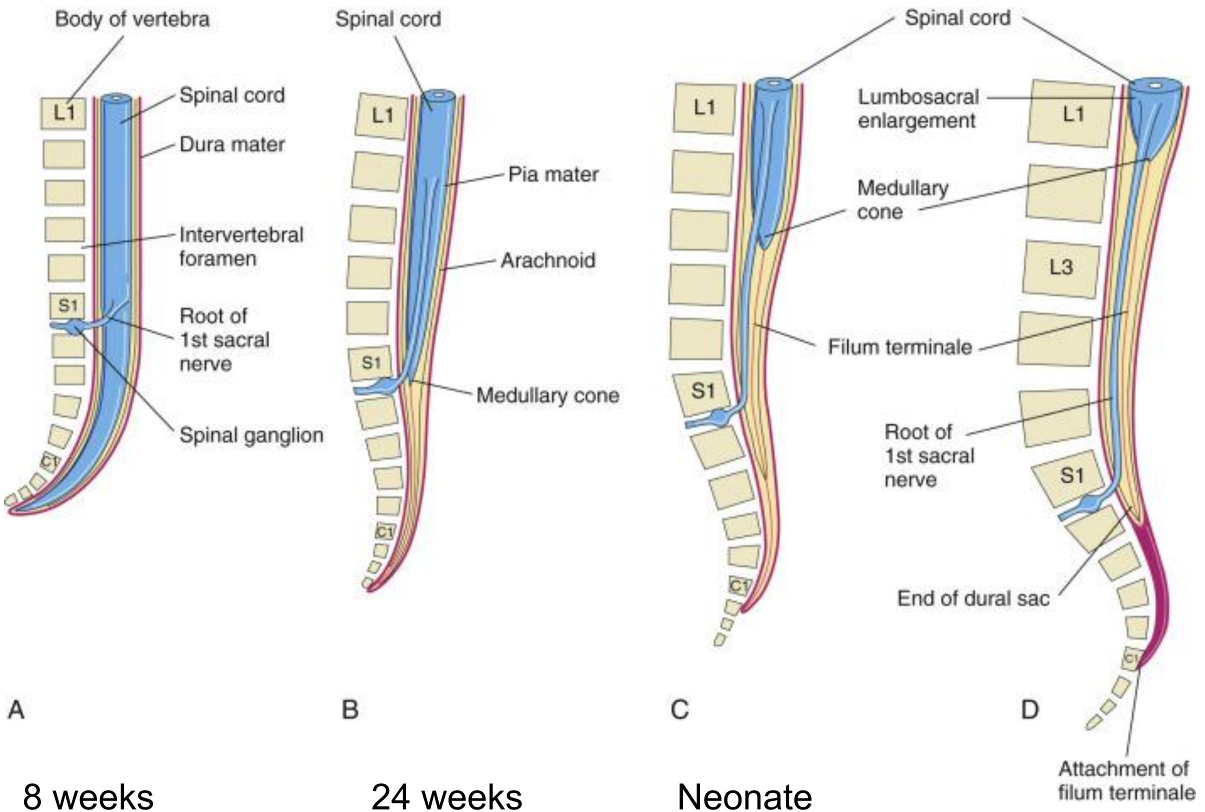
Migrate away from neural tube to form parts of peripheral nervous system:

- Dorsal root ganglia (i.e. somatic sensory neurons)
- Postganglionic autonomic neurons
- Some sensory ganglia of cranial nerves
- Arachnoid and pia mater
- Schwann cells

And also:

- Melanocytes
- Some facial bones
- Odontoblasts→teeth
- Laryngeal and tracheal cartilage
- Aorticopulmonary septum and parts of heart valves

Development of the Spinal Cord



8 weeks

24 weeks

Neonate

Adult

Moore, The Developing Human, Fig. 17.10

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Vertebrae (and surrounding somites) grow faster than the spinal cord.

This shifts the location of the conus medullaris through development.

Note that with spina bifida, the spinal cord can be attached to surrounding tissue, which stretches the spinal cord during growth. This is **tethered cord syndrome**.

Development of the Brain

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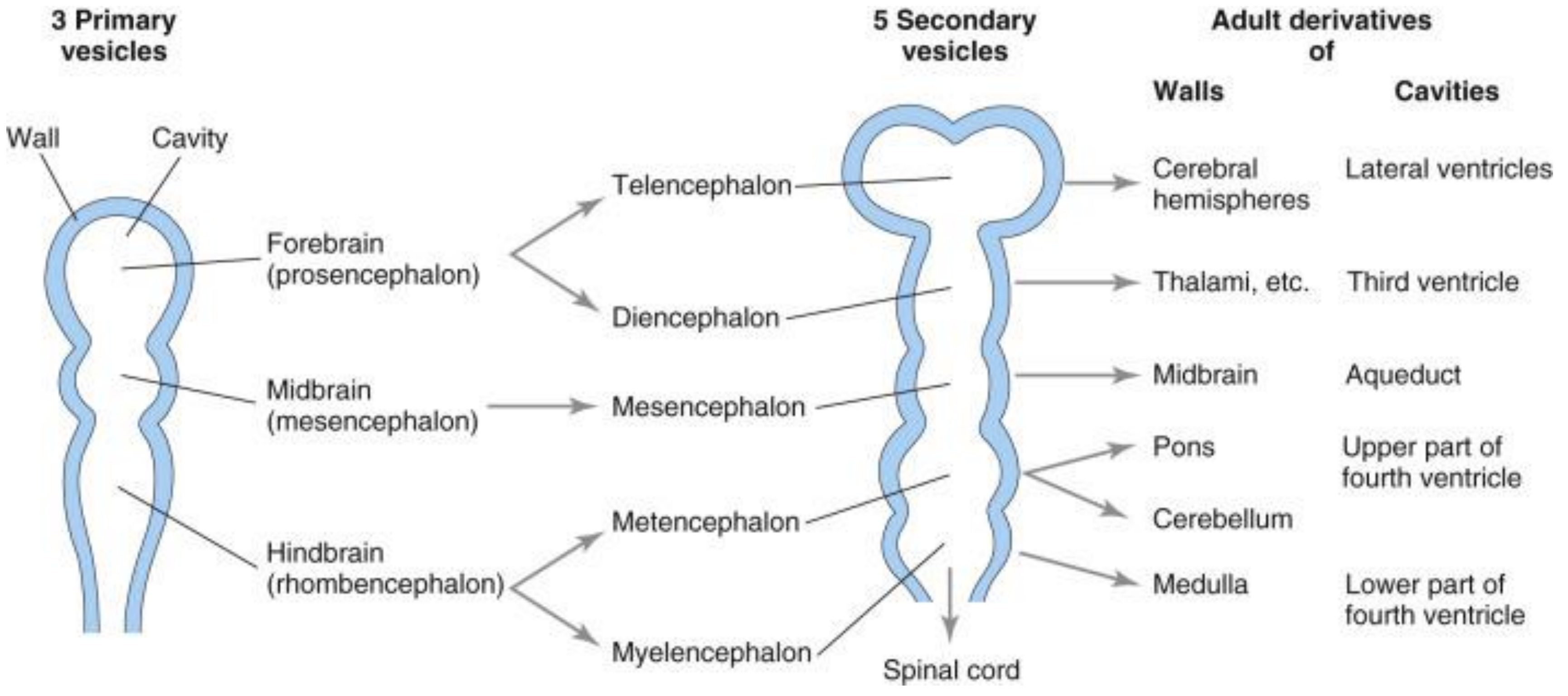

Moore, The Developing Human, Fig. 17.18

Development of the Brain

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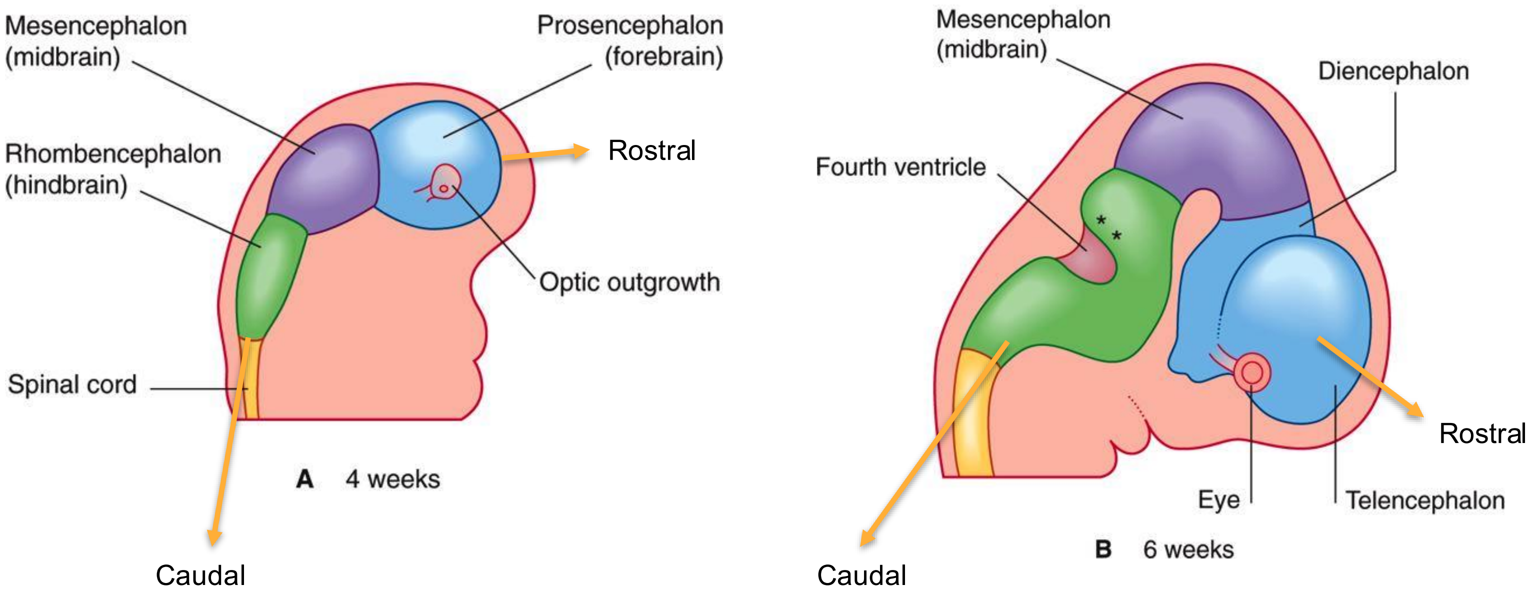
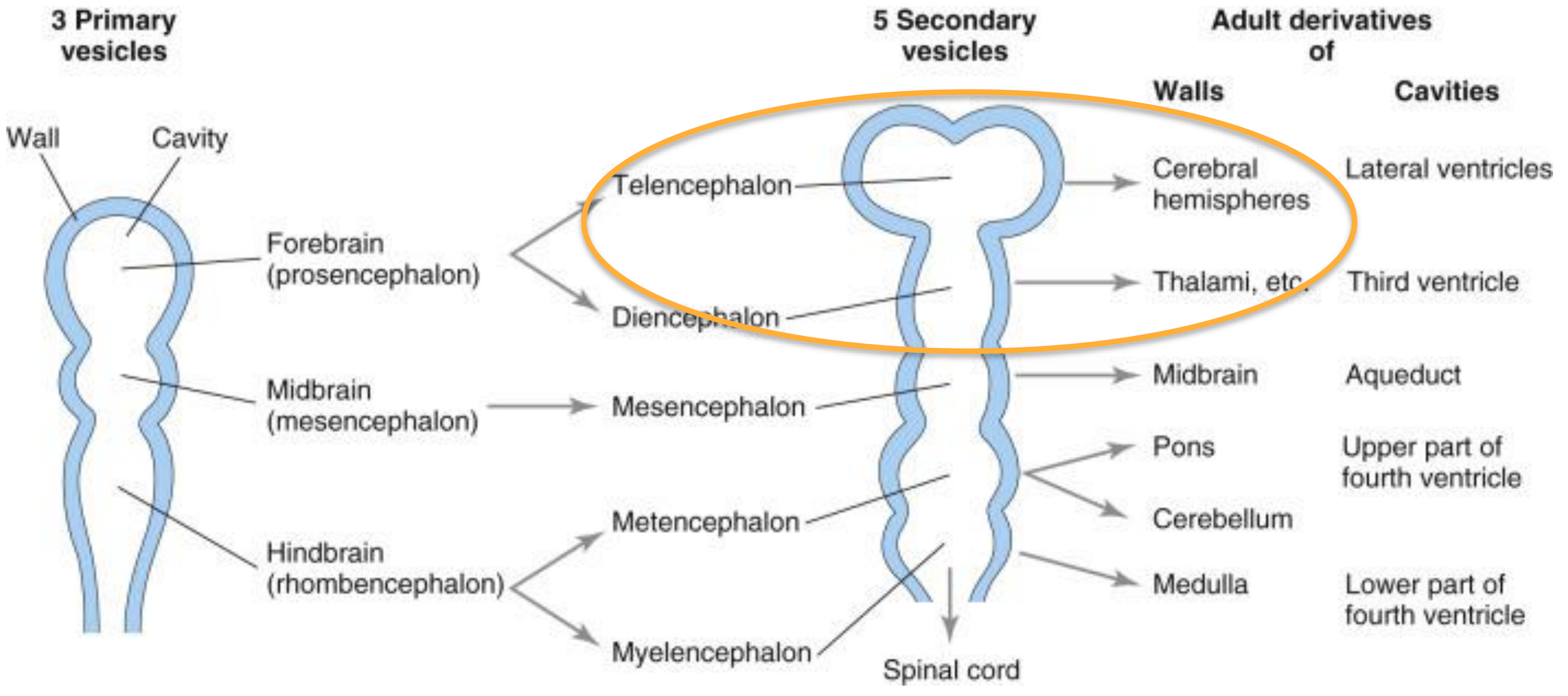


ADULT DERIVATIVES OF THE FOREBRAIN, MIDBRAIN, AND HINDBRAIN

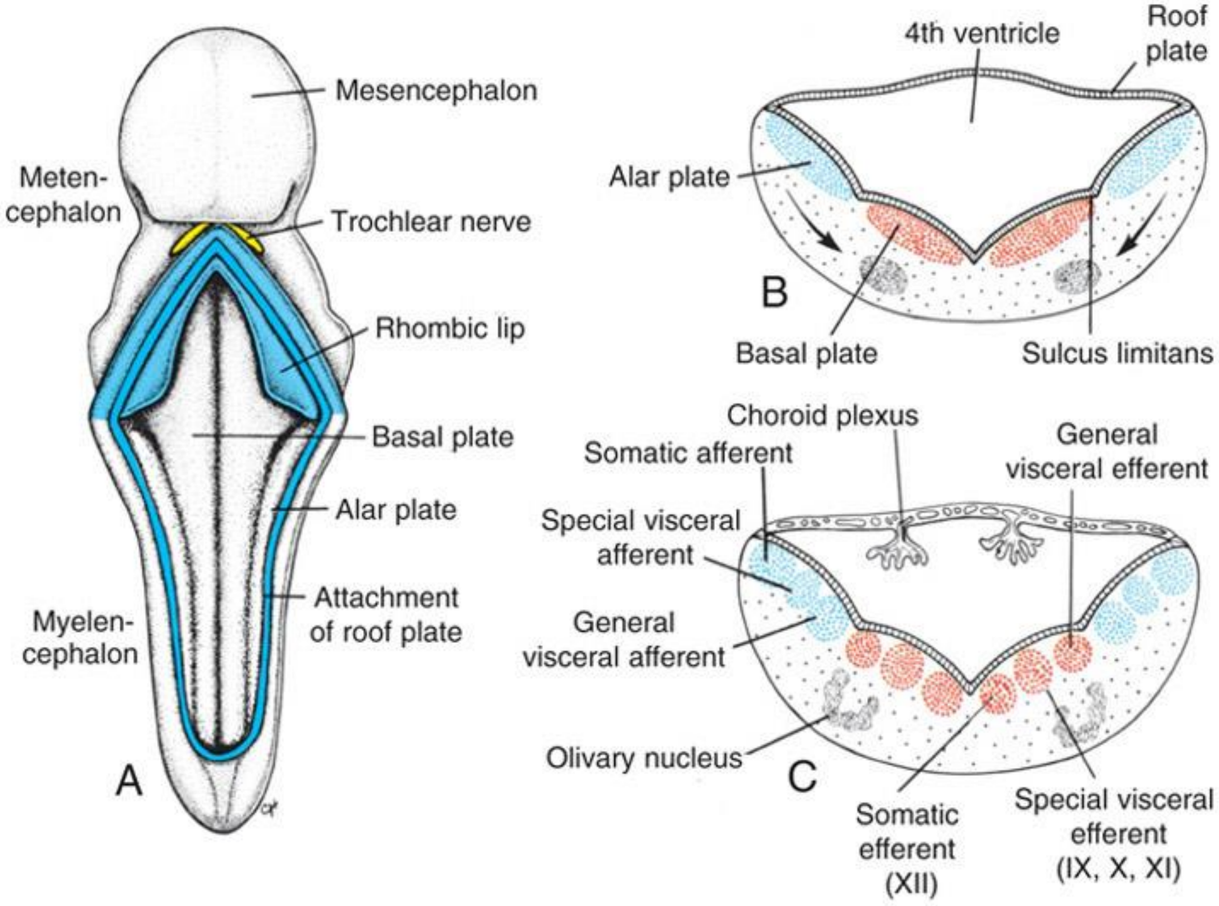
Forebrain	Telencephalon	Cerebral hemispheres (neocortex) Olfactory cortex (paleocortex) Hippocampus (archicortex) Basal ganglia/corpus striatum Lateral and 3rd ventricles	Nerves: Olfactory (I)
	Diencephalon	Optic cup/nerves Thalamus Hypothalamus Mammillary bodies Part of 3rd ventricle	Optic (II)
Midbrain	Mesencephalon	Tectum (superior, inferior colliculi) Cerebral aqueduct Red nucleus Substantia nigra Crus cerebelli	Oculomotor (III) Trochlear (IV)
Hindbrain	Metencephalon	Pons Cerebellum	Trigeminal (V) Abducens(VI) Facial (VII) Acoustic (VIII) Glossopharyngeal (IX) Vagus (X) Hypoglossal (XII)
	Myelencephalon	Medulla oblongata	

Congenital Disorder: Holoprosencephaly



Moore, The Developing Human, Fig. 17.18

Myelencephalon



MYELENCEPHALON

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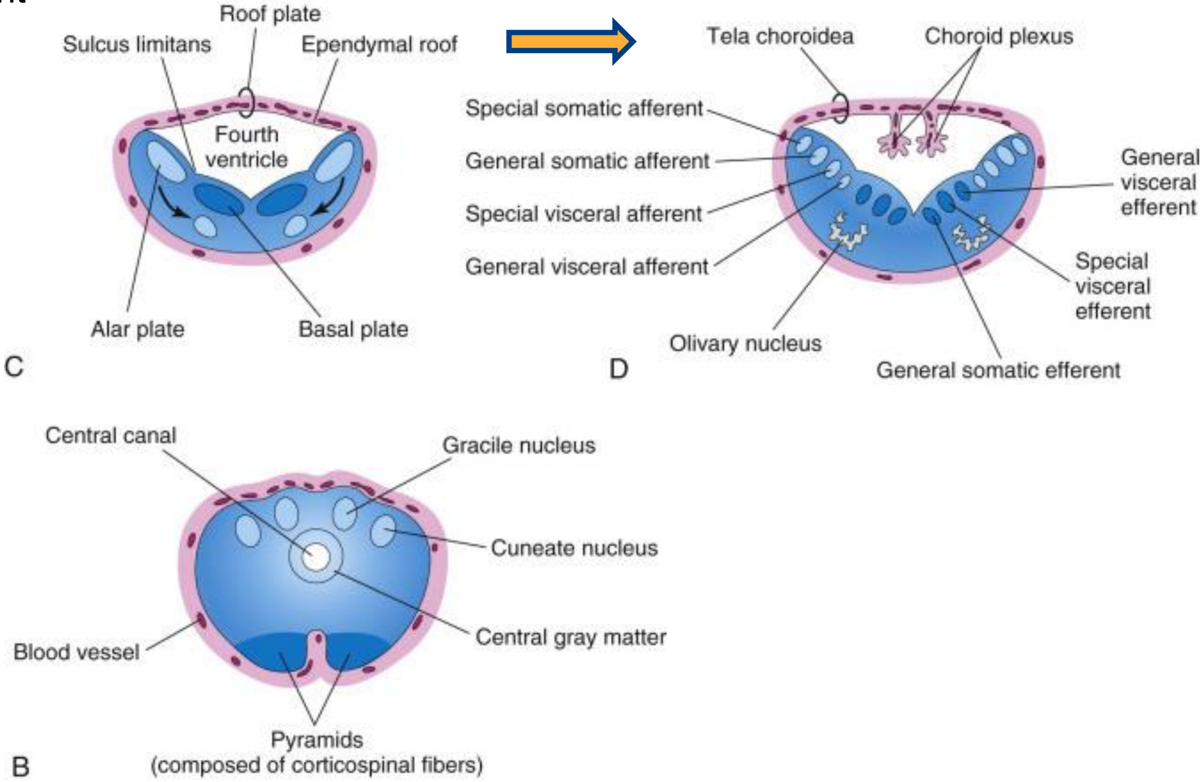
5th week of development

Upper (rostral) medulla

Lower (caudal) medulla



Moore, The Developing Human, Fig. 17.19

FETAL DEVELOPMENT

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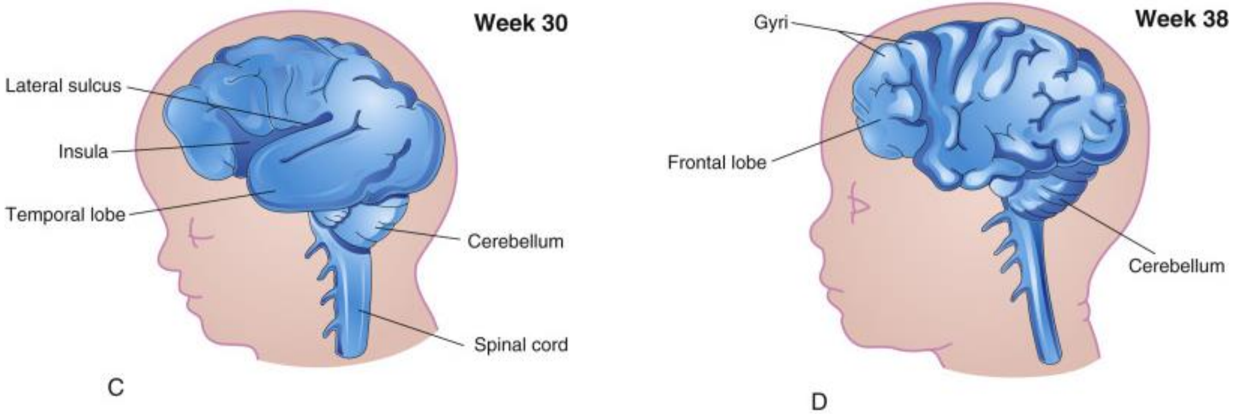
Note that the cortex of the insula becomes covered by the temporal lobe late in fetal develpment



Moore, The Developing Human, Fig. 17.29

CONGENITAL DISORDER: MICROCEPHALY

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Microcephaly—
reduction in neurogenesis. The lack of growth in the brain means the overlying bones of the calvaria also grow slowly

Multiple causes:
Genetic (e.g. autosomal recessive primary microcephaly)

Infectious agents (Zika, rubella, *Toxoplasma gondii*)

Exposure to alcohol (FAS)

Exposure to high amounts of ionizing radiation

Moore, The Developing Human, Fig. 17.36

DEVELOPMENT OF THE VENTRICLES

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Schematic view of ventricles

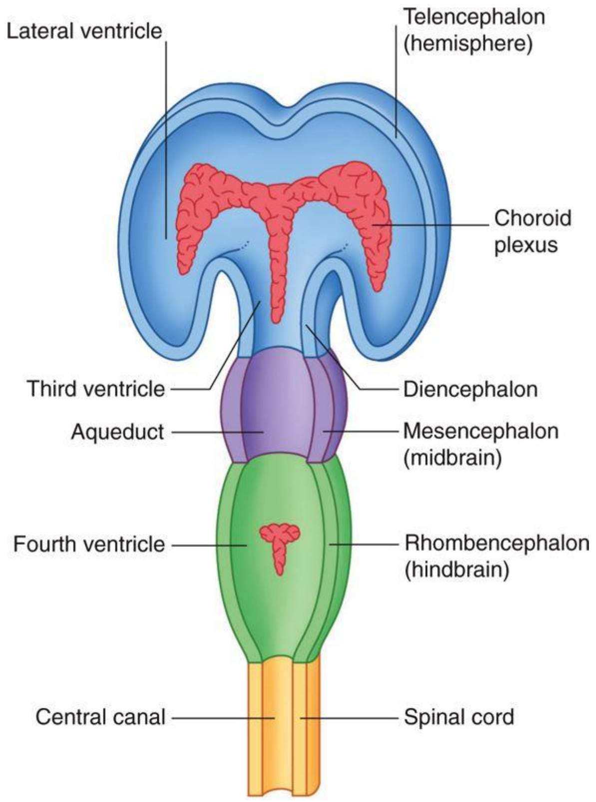

Fig. 1.5, *Clinical Neuroanatomy and Neuroscience, 6th Ed*

Development of the Ventricles

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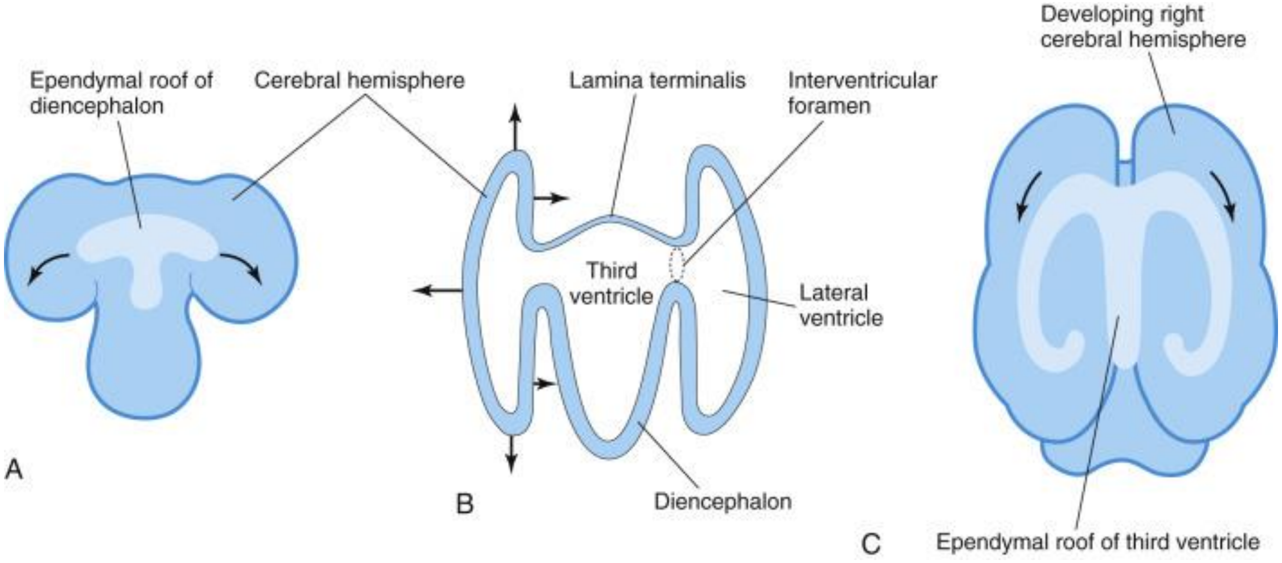

Moore, The Developing Human, Fig. 17.26

DEVELOPMENT OF THE VENTRICLES

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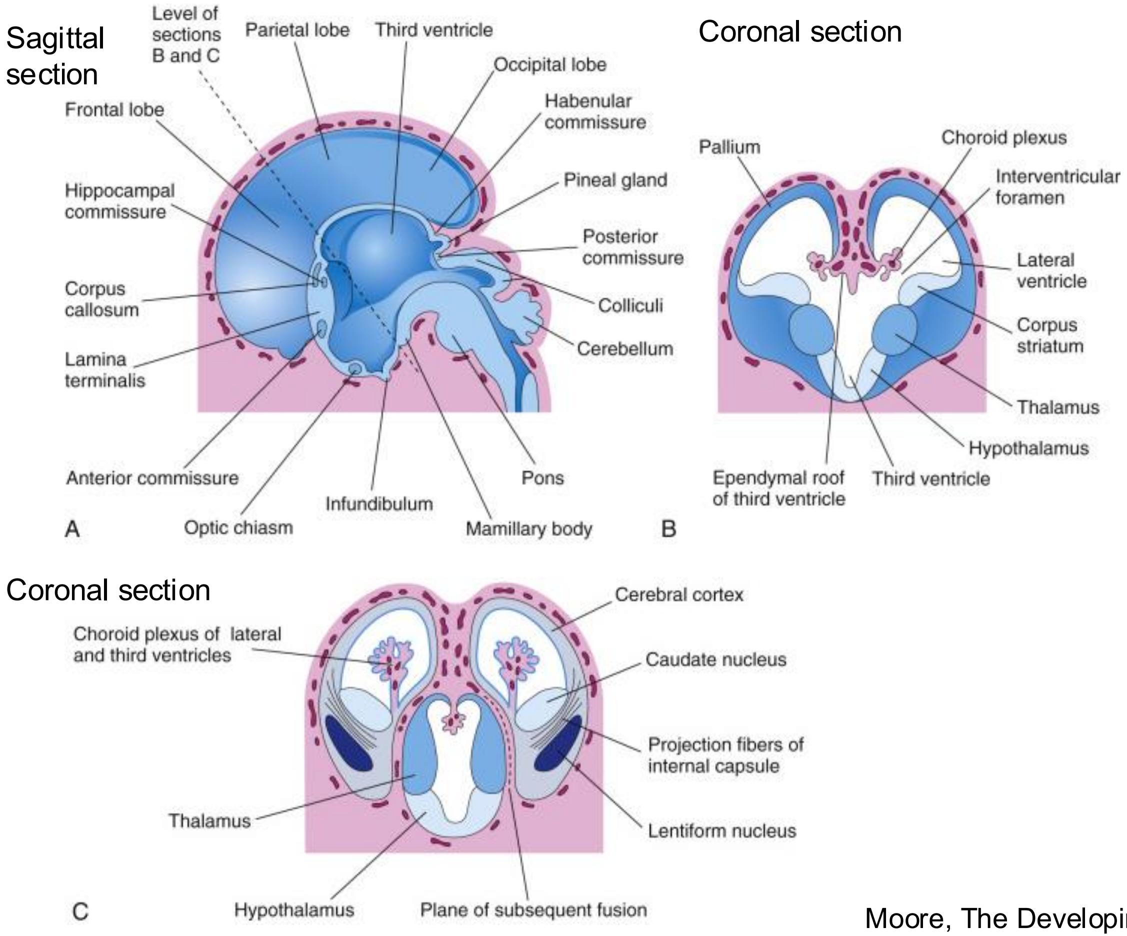


Moore, The Developing Human, Fig. 17.27

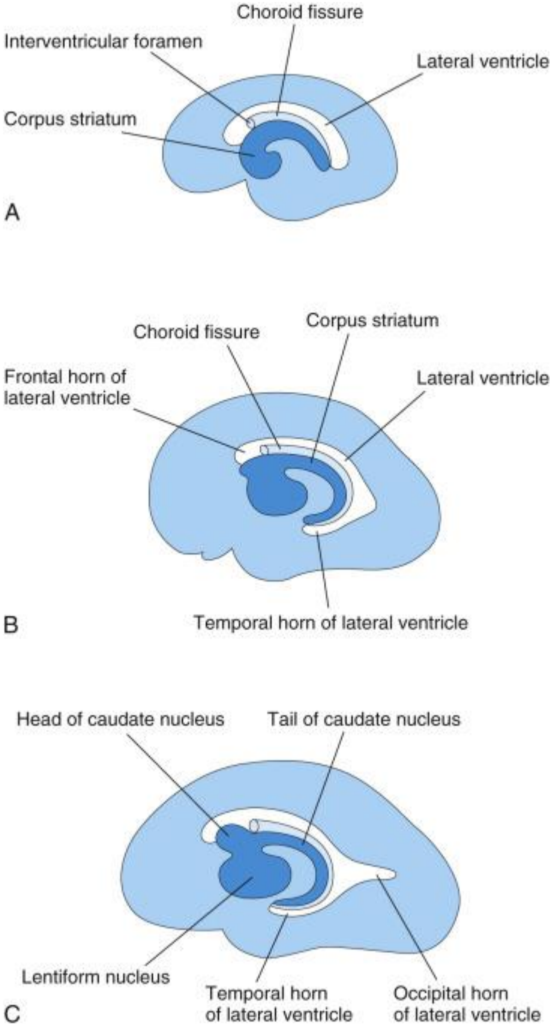
Development of the Lateral Ventricles

Lateral view

13 weeks

21 weeks

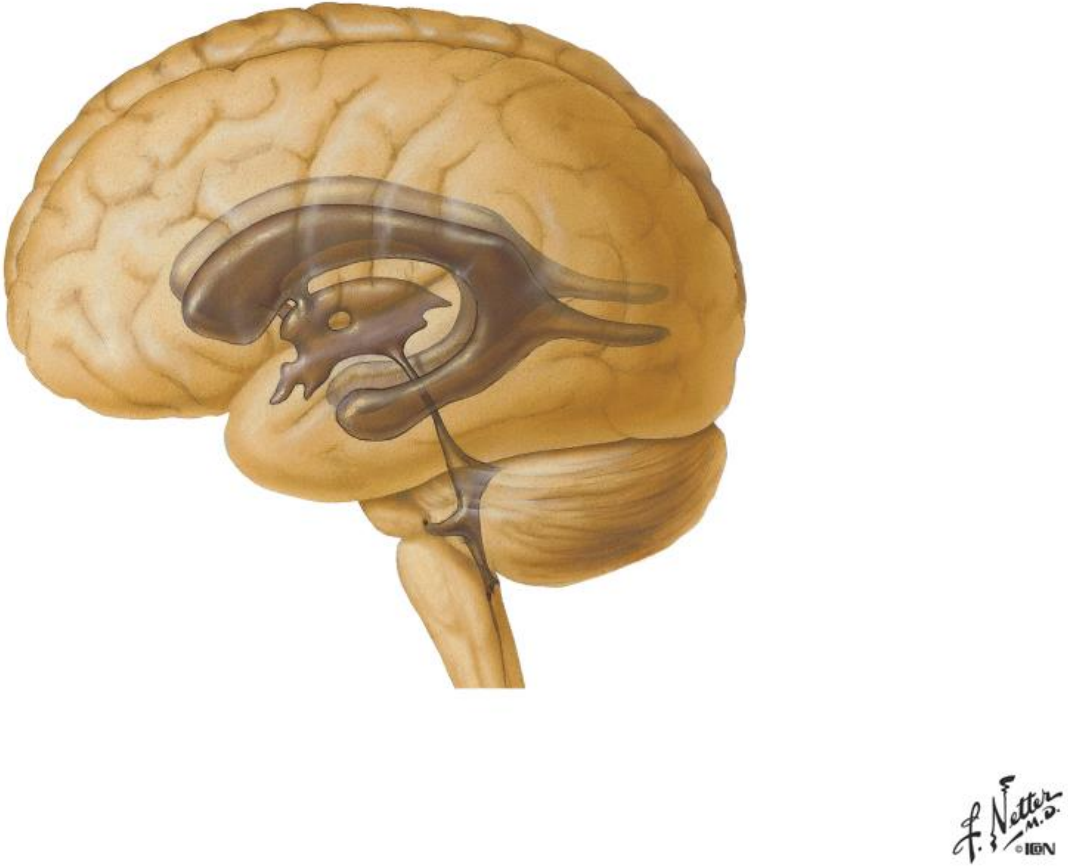
32 weeks



Moore, The Developing Human, Fig. 17.28

Light blue=cerebrum
Dark blue=basal ganglia/nuclei
White=lateral ventricle

FINAL CONFIGURAION OF THE VENTRICLES



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CONGENITAL DISORDER: HYDROCEPHALUS

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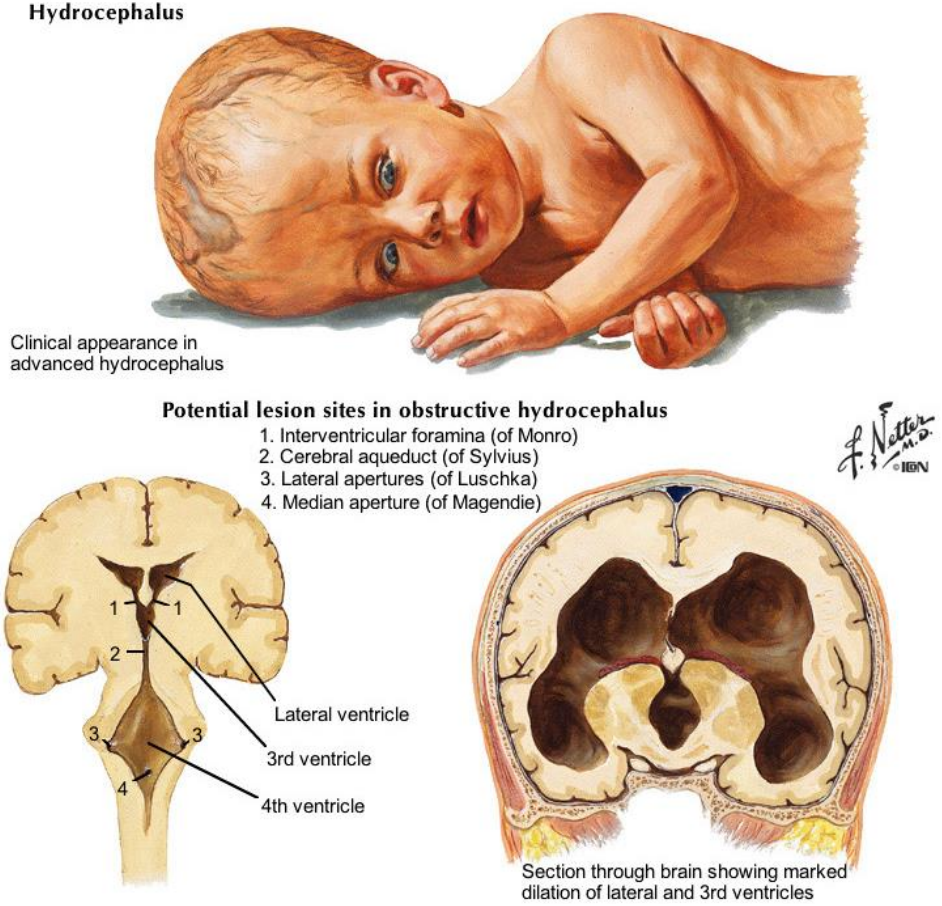

(Non-communicating) Hydrocephalus—
Can occur congenitally when ventricular system is not entirely open.

This happens most commonly in the cerebral aqueduct (congenital aqueductal stenosis).

Signs include large head size with bulging veins and fontanelles, irritability, drowsiness, vomiting, seizures.

Summary Slide

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Disorder	Embryonic structure affected	Result
Spina bifida	Caudal neuropore fails to close	Missing vertebral laminae, possible meningocele, meningomyelocele, or myeloschisis
Anencephaly	Rostral neuropore fails to close	Agenesis of much of brain and overlying bones
Encephalocele	Rostral neuropore fails to close/neural tube fails to separate from ectoderm	Opening in bones of calvaria, possible meningocele, meningoencephalocele, or meningohydroencephalocele
Holoprosencephaly	Prosencephalon fails to divide on midline into paired lobes of telencephalon	Lack of separate left and right cerebral hemispheres, agenesis/hypoplasia of corpus callosum
Dandy-Walker Syndrome	Rhombic lip fails to grow	Agenesis or hypoplasia of cerebellar vermis
Microcephaly	reduction in neurogenesis	Too few/simplified gyri of cerebrum
Hydrocephalus	Blockage/closure in ventricles	Enlarged ventricles, build up of CSF, large head size